

Fuzzing with Code Fragments

Fuzzing with Code Fragments - Christian Holler, Kim Herzig, Andreas Zeller, [usenix.org](https://www.usenix.org).

- Published: date not stated
- Original: <<https://www.usenix.org/conference/usenixsecurity12/technical-sessions/presentation/holler>>
- Preserved from: <https://www.usenix.org/conference/usenixsecurity12/technical-sessions/presentation/holler> (stored) on 2026-08-11
- Capture timestamp: 20151023024250
- Licence: unknown

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Fuzzing with Code Fragments

Fuzz testing is an automated technique providing random data as input to a software system in the hope to expose a vulnerability. In order to be effective, the fuzzed input must be *common enough* to pass elementary consistency checks; a JavaScript interpreter, for instance, would only accept a semantically valid program. On the other hand, the fuzzed input must be *uncommon enough* to trigger exceptional behavior, such as a crash of the interpreter. The *LangFuzz* approach resolves this conflict by using a *grammar* to randomly generate valid programs; the code fragments, however, partially stem from *programs known to have caused invalid behavior before*. LangFuzz is an effective tool for security testing: Applied on the Mozilla JavaScript interpreter, it discovered a total of 105 new severe vulnerabilities within three months of operation (and thus became one of the top security bug bounty collectors within this period); applied on the PHP interpreter, it discovered 18 new defects causing crashes.

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Text of BibTeX entry:

```
@inproceedings {180229, author = {Christian Holler and Kim Herzig and Andreas Zeller}, title = {Fuzzing with Code Fragments}, booktitle = {Presented as part of the 21st USENIX Security Symposium (USENIX Security 12)}, year = {2012}, isbn = {978-931971-95-9}, address = {Bellevue, WA}, pages = {445--458}, url = {https://www.usenix.org/conference/usenixsecurity12/technical-sessions/presentation/holler}, publisher = {USENIX}, } <br><a href="/biblio/export/bibtex/180229">Download</a>
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Abstract:

Fuzz testing is an automated technique providing random data as input to a software system in the hope to expose a vulnerability. In order to be effective, the fuzzed input must be *common enough* to pass elementary consistency checks; a JavaScript interpreter, for instance, would only accept a semantically valid program. On the other hand, the fuzzed input must be *uncommon enough* to trigger exceptional behavior, such as a crash of the interpreter. The *LangFuzz* approach resolves this conflict by using a *grammar* to randomly generate valid programs; the code fragments, however, partially stem from *programs known to have caused invalid behavior before*. LangFuzz is an effective tool for security testing: Applied on the Mozilla JavaScript interpreter, it discovered a total of 105 new severe vulnerabilities within three months of operation (and thus became one of the top security bug bounty collectors within this period); applied on the PHP interpreter, it discovered 18 new defects causing crashes.

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